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Mechanisms and Stress Evolution of Hydraulic Fracture Interacting with Cavity in Carbonate Reservoirs Using the Displacement Discontinuity Method

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Abstract

This study investigates the mechanisms and stress evolution of hydraulic fractures interacting with natural cavities in ultra-deep carbonate formations. Focusing on the Pengshen 6 well at 9026 m depth in Sichuan Basin, China, it aims to understand how cavity presence alters fracture propagation behavior and stress redistribution, thereby improving fracture stimulation strategies in complex reservoir systems.

A physics-based simulation framework is developed using the Displacement Discontinuity Method (DDM) to analyze fracture propagation in the presence of cavities. Fractures and cavities are discretized to capture stress interaction zones, with the fracture front evolution governed by local tensile stress criteria. The model captures real-time pressure changes at the injection point and inside the cavity, integrating Green's function-based stress computations. Field parameters from the Pengshen 6 well are used to simulate realistic stress conditions. A parametric study explores the influence of in-situ stress, Young's modulus, and fluid injection rates on the onset and evolution of fracture-cavity interaction.

Five distinct stages of fracture–cavity interaction are identified: Approach, Contact, Rupture, Initiation, and Propagation. A transient pressure plateau occurs at the injection point during cavity contact, followed by a pressure drop after rupture. Cavity internal pressure shows a delayed rise, indicating fluid penetration post-fracture breakthrough. The stress field near the cavity undergoes notable amplification and local reversal, especially during rupture, affecting subsequent fracture orientation and growth. Higher horizontal minimum stress delays rupture, while higher rock stiffness enhances activation and branching. These results highlight the pivotal role of cavities in altering the fracture path and reshaping the local stress landscape, offering a mechanistic basis for fracture design in deeply buried carbonate reservoirs.

This study provides the first deep-reservoir-focused mechanistic interpretation of fracture–cavity interactions using DDM. By coupling field-calibrated simulations with stress and pressure evolution tracking, it reveals critical insights into stress amplification zones and pressure dynamics around cavities—key factors often overlooked in conventional models. These findings advance the understanding of reservoir stimulation in ultra-deep, cavity-rich carbonate formations.

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Introduction

Carbonate reservoirs, formed predominantly from marine biological deposits, play a critical role in global hydrocarbon production, contributing approximately 60% of the world's oil output (Lucia et al., 1999). These reservoirs are typically characterized by low matrix permeability and the presence of complex fracture–cavity systems where vugs and cavities serve as primary hydrocarbon storage sites (Liu et al., 2020). Despite the widespread application of hydraulic fracturing (HF) to enhance reservoir permeability, the irregular distribution of fractures and cavities in such systems complicates fracture propagation, reduces stimulation efficiency, and poses challenges for optimal treatment design (Ma et al., 2024; Liu et al., 2022). Cavities, in particular, can significantly modify stress fields and fluid pathways, influencing the geometry and effectiveness of induced fractures (Pan et al., 2025). Understanding these interactions is therefore essential for improving recovery in carbonate formations.

Previous studies have investigated the impact of reservoir heterogeneities on hydraulic fracturing using a variety of numerical techniques, including linear elastic finite element methods (LFEM), extended finite element methods (XFEM), and phase field modeling (Cheng et al., 2024; Zhuang et al., 2019). These approaches have successfully simulated fracture dynamics and stress evolution but often neglect the dynamic changes in cavity internal pressure and post-interaction stress redistribution (Luan et al., 2024; Liu et al., 2022). Experimental and numerical analyses have also emphasized the influence of factors such as burial depth, fluid viscosity, and injection rates on fracture connectivity and network complexity (Liu et al., 2022; Luan et al., 2024). However, the specific interactions between hydraulic fractures and cavities—especially the mechanisms governing stress concentration, pressure buildup, and rupture—remain insufficiently understood.

The Displacement Discontinuity Method (DDM) has emerged as an efficient boundary element approach for modeling fractures in elastic media (Crouch, 1983). Owing to its ability to accurately represent stresses in complex geometries, DDM has been widely applied in simulating hydraulic fractures in layered and heterogeneous formations (Wu & Olson, 2015). Yet, its application to fracture–cavity systems in carbonate reservoirs is still limited. Extending DDM to incorporate cavity interactions offers a robust framework for capturing the hydraulic–mechanical coupling processes that control fracture propagation in such heterogeneous environments (Wu et al., 2018).

This study employs DDM to investigate the mechanisms and stress evolution of hydraulic fractures interacting with pre-existing cavities in carbonate reservoirs. The objectives are to: (i) develop a numerical model capable of simulating fracture–cavity interactions, (ii) analyze the pressure and stress field evolution during different interaction stages, and (iii) evaluate the influence of key mechanical parameters such as minimum horizontal principal stress and Young's modulus on fracture–cavity behavior. The insights gained provide a theoretical basis for optimizing hydraulic fracturing designs in complex carbonate reservoirs containing cavities.

Methodology

Displacement Discontinuity Method (DDM)

The Displacement Discontinuity Method (DDM) is a boundary element technique widely used to model fractures in elastic media. In DDM, displacement discontinuities $\mathbf{D}$ on fracture and cavity boundaries are treated as primary unknowns, and the stresses on these boundaries are expressed as linear functions of $\mathbf{D}$ through influence matrices derived from Green's functions. The modeling framework for hydraulic fracture–cavity interactions is schematically illustrated in Fig. 1.

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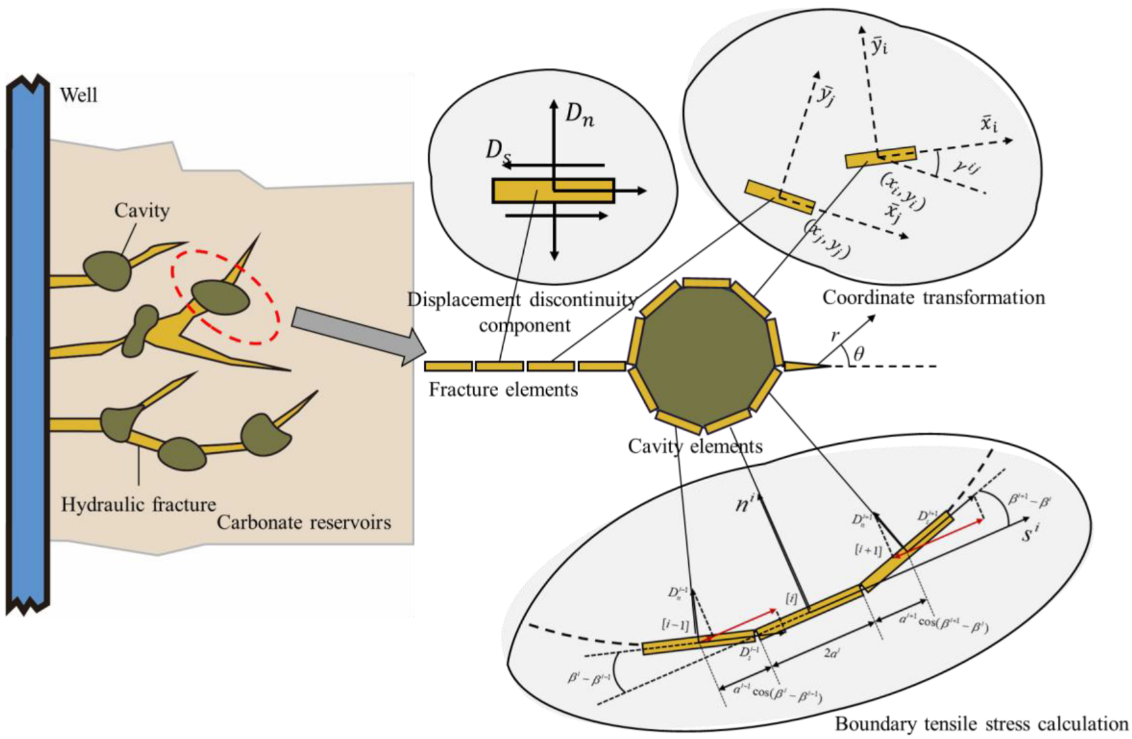

Figure 1—Illustration of the Displacement Discontinuity Method (DDM) for modeling hydraulic fracture–cavity interactions.

For a rock mass containing both fractures and cavity boundaries, the boundary stress is expressed as:

$$\begin{bmatrix}\sigma_n \\ \sigma_s\end{bmatrix}=\begin{bmatrix}A_{nn}^f & A_{nn}^c \\ A_{sn}^f & A_{sn}^c\end{bmatrix}\begin{bmatrix}D_n^f \\ D_n^c\end{bmatrix}+\begin{bmatrix}A_{ns}^f & A_{ns}^c \\ A_{ss}^f & A_{ss}^c\end{bmatrix}\begin{bmatrix}D_s^f \\ D_s^c\end{bmatrix} \tag{1}$$

where D_n and D_s are the normal and shear displacement discontinuities, and A_{nn}^f, A_{nn}^c, etc. are the corresponding influence coefficients.

For fractures subjected to fluid pressure p, the normal stress satisfies σ_n=-p and the shear stress σ_s=0. Substituting these boundary conditions into Eq. (1) gives:

$$\mathbf{A}_n\mathbf{D}_n + \mathbf{A}_s\mathbf{D}_s = -\mathbf{p} \tag{2}$$

Neglecting shear displacement discontinuities ($\mathbf{D}_s$=0), the system reduces to:

$$\mathbf{A}_n\mathbf{w} = -\mathbf{p} \tag{3}$$

where $\mathbf{w}=\mathbf{D}_n$ is the vector of fracture apertures.

Fluid Flow in Fractures and Cavities

Fluid flow within the fracture–cavity network is governed by mass conservation and a Darcy-type momentum balance adapted to fracture geometries. The following assumptions are made:

The fluid is incompressible and Newtonian, with constant dynamic viscosity μ;

Flow within fractures and cavities is dominated by laminar (Poiseuille-type) flow, described by the cubic law;

Flow is one-dimensional along the fracture axis, and cavity pressure is assumed spatially uniform.

Governing equations in fractures. For a fracture segment with aperture *w*, the fluid satisfies the continuity equation:

$$\frac{\partial V_f}{\partial t}+\frac{\partial Q}{\partial s}=Q_0\delta(s) \tag{4}$$

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where $V_f = hw$ is the fluid volume per unit length, $Q(s,t)$ is the volumetric flux along the fracture, and $Q_{0\delta(s)}$ represents a point source or sink.

The momentum equation is given by the cubic law:

$$Q = -\frac{w^3}{12\mu}\frac{\partial p}{\partial s} \tag{5}$$

where $p(s,t)$ is the fluid pressure, and μ is the dynamic viscosity.

Using a one-dimensional finite volume scheme, the fracture is divided into N control volumes. For the i-th control volume:

$$\frac{V_i^1 - V_i^0}{\Delta t} + \frac{Q^1_{i+\frac{1}{2}} - Q^1_{i-\frac{1}{2}}}{\Delta s} = Q_0\delta\left(s_i\right) \tag{6}$$

The flux at the interface $i+1/2$ is approximated by:

$$Q^1_{i+\frac{1}{2}} = -\frac{w^3_{i+\frac{1}{2}}}{12\mu}\frac{P^1_{i+1} - P^1_i}{\Delta s} \tag{7}$$

where $w_{i+1/2}$ is computed by weighted interpolation:

$$w^1_{i+\frac{1}{2}} = \frac{a_{i+1}w^1_{i+1} + a_i w^1_i}{a_{i+1} + a_i} \tag{8}$$

Substituting Eq. (7) into Eq. (6) leads to a tridiagonal system:

$$S_i = C_{i,i-1}P^1_{i-1} + C_{i,i}P^1_i + C_{i,i+1}P^1_{i+1} \tag{9}$$

where the coefficients $C_{i,j}$ depend on w_3, reflecting the coupling between fluid flow and fracture deformation.

Governing equations in cavities. Each cavity is treated as a lumped node with uniform pressure p_c and volume V_c. The mass conservation equation for the k-th cavity is:

$$\sum_{j\in\mathcal{C}^{(k)}} Q_j^{(k)} = \frac{\Delta V_c^{(k)}}{\Delta t} \tag{10}$$

where $Q_j^{(k)}$ is the flux between the cavity and the j-th connected fracture segment, given by:

$$Q_j^{(k)} = -\frac{w_j^3}{12\mu L_j}\left(p_j - p_c^{(k)}\right) \tag{11}$$

Rearranging gives the discrete form:

$$\sum_j \alpha_j^{(k)} p_j - \left(\sum_j \alpha_j^{(k)}\right) p_c^{(k)} = \frac{\Delta V_c^{(k)}}{\Delta t} \quad \text{with} \quad \alpha_j^{(k)} = \frac{w_j^3}{12\mu L_j} \tag{12}$$

The cavity volume increment is obtained from the structural deformation:

$$\Delta V_c^{(k)} = \eta_k h_c^{(k)} \sum_{i=1}^{N_b} \left(\mathbf{u}_i \cdot \mathbf{n}_i\right) L_i \tag{13}$$

where $h_c^{(k)}$ is the characteristic cavity height, $\eta_k \in (0,1]$ is a shape correction factor, and $\mathbf{u}_i{\cdot}\mathbf{n}_i$ is the normal displacement of the boundary.

By assembling the discretized equations for all fracture segments and cavities, the fluid subsystem is expressed as:

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$$\mathbf{C}(\mathbf{w})\mathbf{p} + \mathbf{M}_c\mathbf{p}_c = \mathbf{S}(\mathbf{w}) \tag{14}$$

$$\mathbf{M}_c^{\top}\mathbf{p} - \mathbf{C}_c\mathbf{p}_c = \mathbf{b}_c \tag{15}$$

where: $\mathbf{C}(\mathbf{w})$ is the transmissibility matrix for fractures, $\mathbf{M}_c$ is the fracture–cavity connectivity matrix, $\mathbf{C}_c$ is a diagonal matrix with cavity conductances, $\mathbf{b}_c$ is the vector of cavity volume change rates.

Fully Coupled Hydro-Mechanical Model

The interaction between fluid and solid is **two-way**: fluid pressure p applies a normal stress on fracture walls, and fracture aperture w influences hydraulic transmissivity w_3.

Combining the solid equation (DDM) and the fluid equations, the coupled unknown vector is:

$$\mathbf{x} = \begin{bmatrix} \mathbf{p} \\ \mathbf{p}_c \\ \mathbf{W} \end{bmatrix} \tag{16}$$

The global coupled system is formulated as:

$$\begin{bmatrix} \mathbf{C}(\mathbf{w}) & \mathbf{M}_c & \mathbf{0} \\ \mathbf{M}_c^{\top} & -\mathbf{C}_c & \mathbf{0} \\ \mathbf{I} & \mathbf{0} & \mathbf{A}_n \end{bmatrix} \begin{bmatrix} \mathbf{p} \\ \mathbf{p}_c \\ \mathbf{w} \end{bmatrix} = \begin{bmatrix} \mathbf{S}(\mathbf{w}) \\ \mathbf{b}_c \\ \mathbf{0} \end{bmatrix} \tag{17}$$

where: $\mathbf{A}_n$ is the DDM matrix relating fracture aperture to pressure, $\mathbf{C}(\mathbf{w})$ is the nonlinear fluid conductivity matrix, $\mathbf{M}_c$, $\mathbf{C}_c$ define the cavity–fracture hydraulic connections.

The coupled nonlinear problem is solved using a fully implicit scheme:

Initialization: set initial $\mathbf{w}^0$, $\mathbf{p}^0$, $\mathbf{p}_c^0$;

Assembly: build matrices $\mathbf{C}(\mathbf{w})$, $\mathbf{M}_c$, and $\mathbf{A}_n$;

Solve: use Newton–Raphson or block-preconditioned iterative methods to solve Eq. (16);

Update: update $\mathbf{w}$, $\mathbf{p}$, and $\mathbf{p}_c$ until convergence $\| \mathbf{x}^{n+1} - \mathbf{x}^n \| < \epsilon$.

Figure 1 Illustration of the Displacement Discontinuity Method (DDM) for modelling hydraulic fracture-cavity interactions.

Results and Discussions

Accuracy verification during fracture propagation

The accuracy of the proposed model is validated by comparing its results with the analytical Khristianovic-Geertsma-de Klerk (KGD) model, which describes hydraulic fracture propagation under plane-strain conditions. The KGD model provides analytical solutions for fracture length $L_f(t)$, net pressure $p_{net}(t)$, and fracture width $W(t)$, given by:

$$L_f(t) = 0.529\left(\frac{q^3 E'}{\nu h^3}\right)^{\frac{1}{6}} t^{\frac{2}{3}} \tag{18}$$

$$p_{net}(t) = 1.09\left(E'^2 \nu\right)^{\frac{1}{3}} t^{-\frac{1}{3}} \tag{19}$$

$$W(t) = 2.36\left(\frac{\nu q^3}{E' h^3}\right)^{\frac{1}{6}} t^{\frac{1}{3}} \tag{20}$$

For model validation, the following parameters were used: Young's modulus E' = 45 GPa, fracturing fluid viscosity ν = 100 mPa·s, initial fracture length $L_f(0)$ = 60 m, relaxation ratio ξ = 0.2, injection flow rate q = 3.18 m³/min, and minimum horizontal principal stress σ_h = 30.68 MPa. These values represent typical reservoir conditions and ensure a meaningful comparison between numerical and analytical results.

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The validity of the proposed model was assessed by comparing its predictions with the analytical Khristianovic–Geertsma–de Klerk (KGD) model under identical reservoir and operational parameters. Fig. 2 illustrates the temporal evolution of the key fracture parameters obtained from both approaches. As shown in Fig. 2(a), the fracture half-length predicted by the proposed model closely matches the KGD analytical solution, demonstrating the model's ability to capture fracture propagation dynamics. Similarly, the pressure evolution inside the fracture (Fig. 2(b)) exhibits excellent agreement, particularly at later stages where both curves converge. The fracture width (Fig. 2(c)) and fracture volume (Fig. 2(d)) also align well with the analytical results, with only minor deviations observed during the early propagation phase. These comparisons confirm that the proposed model can reliably reproduce the fundamental characteristics of hydraulic fracture growth described by the KGD model.

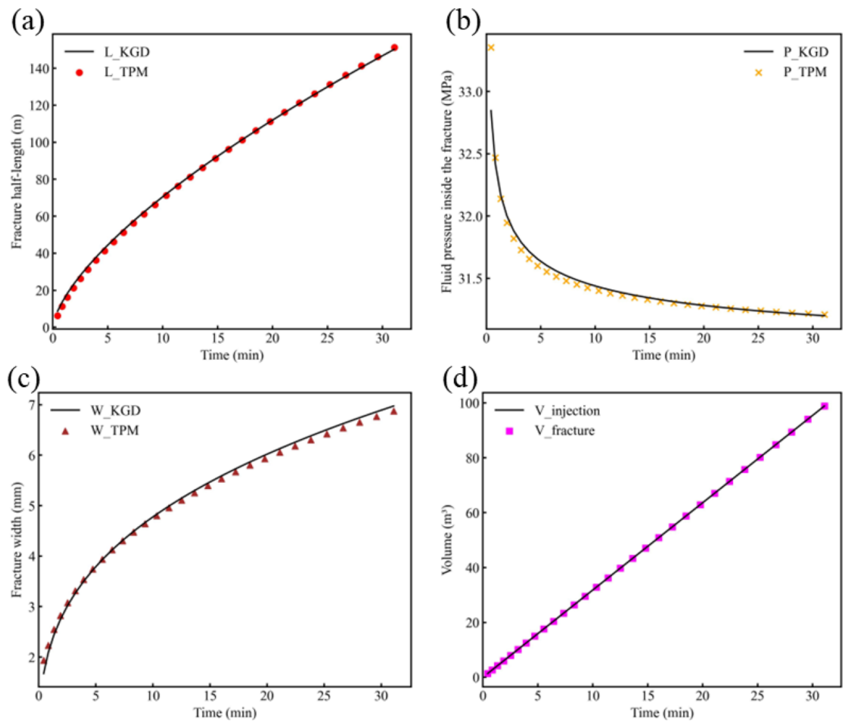

Figure 2—Comparison between the proposed model and the analytical KGD model in terms of (a) fracture half-length, (b) fluid pressure inside the fracture, (c) fracture width, and (d) fracture volume during hydraulic fracturing.

Verification of Stress Field around a Cavity

The accuracy of the proposed discontinuous displacement method (DDM) in predicting the stress distribution around a cavity was validated by comparing the numerical results with the analytical solution for a pressurized circular hole in an infinite elastic medium. The analytical stress components in polar coordinates are expressed as:

$$\sigma_{rr}=\frac{\sigma_{xx}+\sigma_{yy}}{2}\left(1-\frac{a^2}{r^2}\right)+\left(1-4\frac{a^2}{r^2}+3\frac{a^4}{r^4}\right)\left[\frac{\sigma_{xx}-\sigma_{yy}}{2}\cos(2\theta)+\tau_{xy}\sin(2\theta)\right] \tag{21}$$

$$\sigma_{\theta\theta}=\frac{\sigma_{xx}+\sigma_{yy}}{2}\left(1+\frac{a^2}{r^2}\right)-\left(1+3\frac{a^4}{r^4}\right)\left[\frac{\sigma_{xx}-\sigma_{yy}}{2}\cos(2\theta)+\tau_{xy}\sin(2\theta)\right] \tag{22}$$

$$\tau_{r\theta}=\left(1+2\frac{a^2}{r^2}-3\frac{a^4}{r^4}\right)\left[\frac{\sigma_{yy}-\sigma_{xx}}{2}\sin(2\theta)+\tau_{xy}\cos(2\theta)\right] \tag{23}$$

Assuming an arbitrary far-field stress state with σ_{xx} = 20 MPa, σ_{yy} = 10 MPa, and τ_{xy} = 5 MPa, the DDM was applied to compute the boundary stresses at the cavity wall ($r = a$) and the stresses along a specified line. The calculated results were compared against the analytical solution to verify the accuracy of the proposed model in simulating the stress field around a cavity.

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The comparison between the numerical and analytical stress components is shown in Fig. 3, where the numerical results (hollow markers) closely match the analytical solutions (solid lines). This excellent agreement demonstrates the capability of the proposed DDM to accurately capture both the boundary stress at the cavity wall and the stress distribution in the surrounding formation.

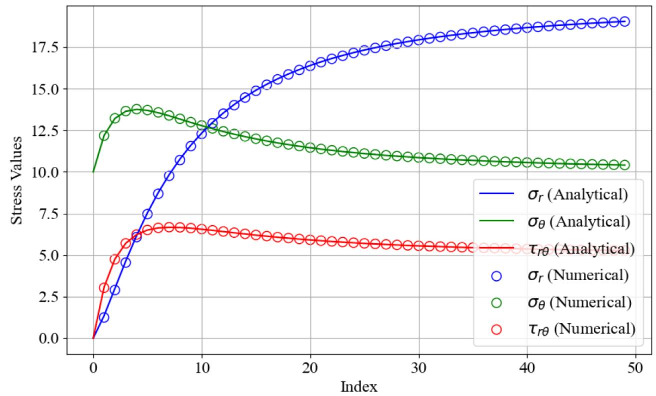

Figure 3—Comparison of stress components σ_r, σ_θ, and $\tau_{r\theta}$ along a specified line between the numerical results from the discontinuous displacement method (DDM) and the analytical solution for a pressurized circular cavity.

Simulation Parameters

The hydraulic fracturing simulation was performed using parameters representative of typical reservoir conditions, as summarized in Table 1. The rock was modeled as an elastic medium with a Young's modulus of 30 GPa and a Poisson's ratio of 0.2. The tensile strength was set to 80 MPa, while the in-situ stresses consisted of a maximum horizontal stress of 35 MPa and a minimum horizontal stress of 30 MPa. A constant injection rate of 6 m³/min was applied using a Newtonian fluid with a viscosity of 100 mPa·s. The fracture height was fixed at 60 m. The model configuration included an initial fracture located at the origin (0,0) and a circular cavity with a diameter of 1 m centered at (4,0). The fracture propagation was discretized using displacement discontinuity elements with a length of 0.2 m, enabling accurate representation of the fracture–cavity interaction.

Table 1—Simulation Parameters

Parameter	Symbol	Value	Unit	Description
Young's modulus	E	30	GPa	Rock elastic modulus
Poisson's ratio	ν	0.2	–	Material's lateral strain ratio
Tensile strength	σ_T	80	MPa	Rock tensile failure threshold
Max horizontal principal stress	σ_H	35	MPa	Regional maximum in-situ stress
Min horizontal principal stress	σ_h	30	MPa	Regional minimum in-situ stress
Fluid injection rate	Q_0	6	m³/min	Constant surface pumping rate
Fluid dynamic viscosity	μ	100	mPa·s	Newtonian fluid viscosity
Fracture height	H	60	m	Vertical extent of the fracture
Initial fracture location	–	(0, 0)	m	Coordinates of fracture origin
Cavity center location	–	(4, 0)	m	Coordinates of circular cavity center
Cavity diameter	–	1	m	Diameter of pre-existing circular cavity
DDM element length	L	0.2	m	Length of each displacement discontinuity element

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Interaction Mechanism between Hydraulic Fracture and Cavity

The interaction process between the hydraulic fracture and the pre-existing cavity is characterized by a series of distinct mechanical and hydraulic responses, as illustrated in Fig. 4. The simulation results reveal a five-stage evolution—approach, intersect, rupture, initiate, and propagate—each associated with characteristic variations in injection pressure, cavity pressure, and local stress distribution.

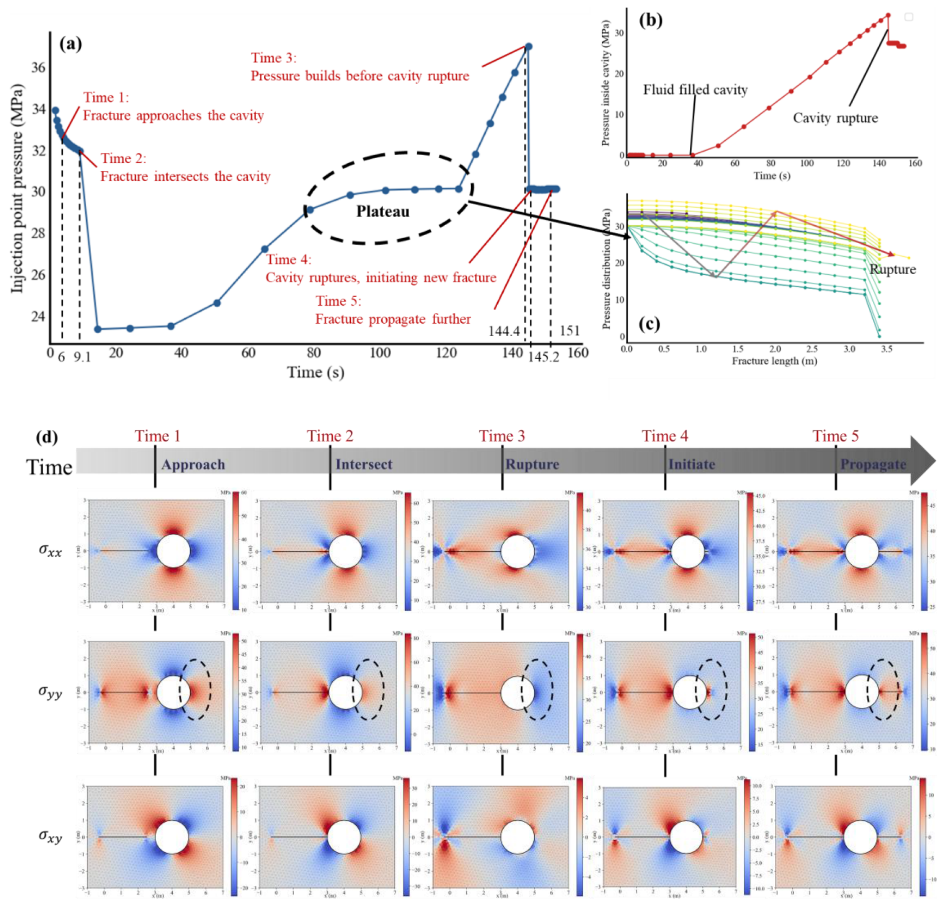

Figure 4—Hydraulic fracture interaction with a circular cavity. (a) Injection point pressure evolution showing five distinct stages: approach, intersect, rupture, initiate, and propagate. (b) Internal cavity pressure response during fracture–cavity interaction. (c) Pressure distribution along the fracture length at different times. (d) Evolution of the stress field around the fracture–cavity system corresponding to the five stages.

Fig. 4(a) displays the injection point pressure history. Initially, as the fracture propagates towards the cavity (Time 1: Approach), the pressure decreases due to fracture extension into the unstressed medium. Once the fracture tip reaches the cavity wall (Time 2: Intersect), fluid begins to penetrate the cavity, leading to a redistribution of the fluid pressure and an initial plateau in the injection pressure curve. This plateau reflects the temporary stabilization of the system as the cavity fills. Subsequently, before rupture occurs, pressure builds up significantly (Time 3: Pressure build-up), indicating resistance from the cavity wall. Upon reaching the critical stress condition, the cavity ruptures (Time 4: Rupture), leading to a sharp pressure drop and the initiation of a new fracture from the cavity boundary. Finally, in the Time 5: Propagate stage, the newly formed fracture continues to extend, and the pressure stabilizes at a lower level than the pre-rupture peak.

Fig. 4(b) further confirms this process by showing the internal cavity pressure response. The cavity remains unpressurized until the main fracture intersects it. Once connected, fluid rapidly fills the cavity, causing the internal pressure to rise almost linearly until rupture. After rupture, the pressure within the

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cavity declines abruptly as the newly created fracture provides an additional fluid pathway, relieving the built-up pressure.

The fracture pressure distribution along its length, shown in Fig. 4(c), evolves dynamically during the interaction. Prior to cavity contact, the pressure decreases monotonically from the injection point to the fracture tip, consistent with classical hydraulic fracture propagation. After the fracture intersects the cavity, a localized pressure increase is observed near the cavity location, indicating fluid accumulation. This is followed by a sharp pressure drop coinciding with cavity rupture, after which the pressure distribution re-establishes a gradient along the newly propagated fracture path.

Sensitivity Analysis of Mechanical Parameters on Fracture–Cavity Interaction

The effects of mechanical parameters on the hydraulic fracture–cavity interaction were examined by conducting a sensitivity analysis on the minimum horizontal principal stress (σ_h) and Young's modulus (E), as illustrated in Fig. 5. Variations in these parameters significantly alter the pressure response, rupture timing, and fracture propagation behavior.

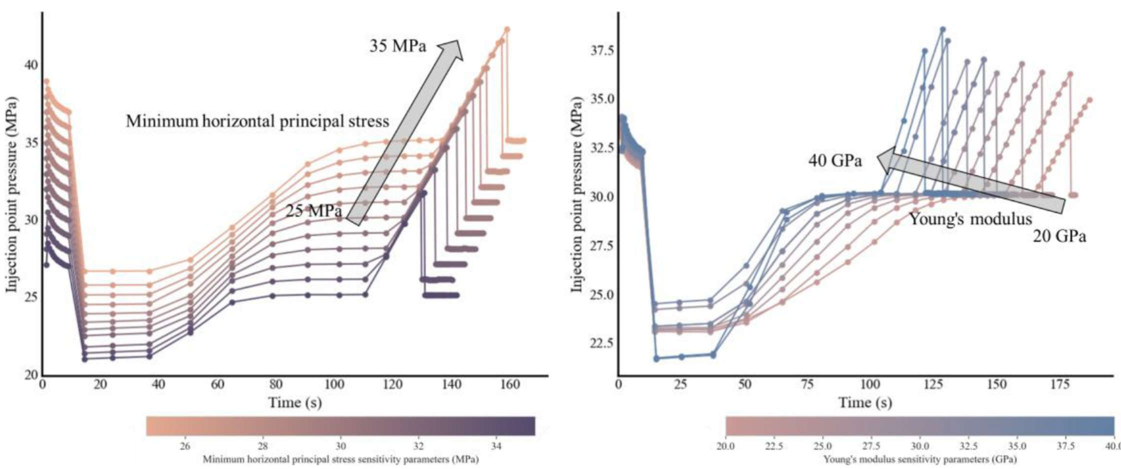

Figure 5—Sensitivity of injection point pressure to (a) minimum horizontal principal stress (σ_h) and (b) Young's modulus (E) during hydraulic fracture–cavity interaction.

In Fig. 5(a), the influence of (σ_h) is evident. When σ_h increases from 25 MPa to 35 MPa, the injection pressure curve rises substantially across all stages. Higher $\sigma_{(h}$ delays the cavity rupture event, as reflected by the rightward shift of the rupture pressure peak. Moreover, under elevated σ_h, the initiation of a new fracture is suppressed, demonstrating that higher in-situ compressive stress enhances the mechanical stability of the cavity and reduces its susceptibility to failure.

Conversely, Fig. 5(b) shows the impact of Young's modulus (E) on the interaction process. Increasing E from 20 GPa to 40 GPa elevates the propagation pressure and slightly increases the rupture pressure. However, unlike σ_h, a stiffer rock (higher E) accelerates the rupture process, shortening the time to cavity failure and promoting the initiation of new fractures. This behavior is attributed to the increased ability of stiff formations to transmit stress concentrations efficiently to the cavity boundary, facilitating early rupture and subsequent fracture formation.

Conclusion

This study employed the Displacement Discontinuity Method (DDM) to investigate hydraulic fracture–cavity interactions in carbonate reservoirs. The results demonstrate that pre-existing cavities significantly influence fracture propagation, leading to a characteristic five-stage evolution – approach, intersect, rupture, initiate, and propagate—accompanied by distinct pressure plateaus and local stress reversals. These

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findings highlight the strong mechanical–hydraulic coupling between fractures and cavities and reveal that cavities act not only as fluid storage zones but also as stress concentrators that can trigger secondary fracture initiation. Sensitivity analyses further indicate that a higher minimum horizontal principal stress delays cavity rupture and suppresses secondary fracture formation, whereas a higher Young's modulus promotes earlier cavity activation and fracture branching. Such observations provide theoretical guidance for optimizing hydraulic fracturing designs in carbonate reservoirs with complex cavity systems, particularly in balancing stimulation efficiency with reservoir stability.

Despite the promising results, certain limitations remain in this study. The present model assumes linear elasticity of the rock matrix and uniform cavity geometry, which may oversimplify actual reservoir conditions where heterogeneity, anisotropy, and irregular cavity shapes play a crucial role. Additionally, thermal effects, multiphase fluid interactions, and proppant transport were not incorporated into the current framework, potentially limiting the direct applicability to field-scale treatments.

Future work should therefore focus on extending the model to include nonlinear rock behavior, heterogeneous and anisotropic material properties, and more realistic cavity morphologies. Incorporating thermal–hydro–mechanical–chemical (THMC) coupling and proppant transport mechanisms would also enhance the model's applicability for field-scale hydraulic fracturing design. Furthermore, validation using laboratory experiments and field data is necessary to verify the predictive capability of the model and to establish practical guidelines for stimulation optimization in cavity-bearing carbonate reservoirs.

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